

THE REVERSAL CURSE: LLMs TRAINED ON “A IS B” FAIL TO LEARN “B IS A”

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TL;DR

This paper shows a sharp failure mode in autoregressive language models (models that predict text one token at a time): if you train them on facts written as “A is B,” they often learn to answer questions in that same direction but fail almost completely to infer or retrieve the reversed form “B is A,” even when the reversal is logically equivalent.

Prerequisites

- autoregressive language model (a model that generates text by predicting the next word or token from the previous ones)
- fine-tuning (taking a pretrained model and training it a bit more on a smaller task-specific dataset)
- likelihood / conditional probability (how much probability the model assigns to one answer given a prompt)
- generalization (when a model learns a pattern that works on new examples, not just the exact training text)
- in-context learning (getting a model to use information placed directly in the prompt, without changing its weights)
- knowledge graph (a structured database of facts where entities and relations are stored explicitly)

The Big Idea

The paper is about a very basic-seeming question: if a language model learns the fact “A is B,” does it also learn “B is A” when those two statements mean the same thing? You would expect yes. Humans do this without thinking. If you learn “Valentina Tereshkova was the first woman in space,” you can answer both “Who was Valentina Tereshkova?” and “Who was the first woman in space?” The paper shows that large language models often do not do this after training. That is the “Reversal Curse.”

What makes this interesting is that the failure is not subtle. The authors fine-tune models on synthetic facts like “Daphne Barrington is the director of ‘A Journey Through Time.’” Then they test both directions. The model often succeeds when the prompt matches the training order, but in the reversed direction it can drop to chance. In plain English: the model seems to memorize a one-way association, not the underlying identity. It is like learning that a phone number belongs to Alice, but then failing to recognize Alice when someone gives you the phone number.

This matters because people often talk about LLMs as if they build an internal world model or a flexible store of facts. This paper says: not so fast. At least for this kind of factual learning, the model behaves more like a directional text-completion machine than a system that automatically closes simple logical

loops. That is a big deal if you want to use LLMs as knowledge bases, or if you think scaling alone will make these basic reasoning gaps disappear.

The paper's core claim is also nicely targeted. They are not saying LLMs cannot do reversal at all. In fact, if you put "A is B" directly in the context window, strong models can often infer "B is A." So the issue is not raw reasoning ability at test time. The issue is what gets absorbed during training. That distinction is important, and I think it is the sharpest point in the paper.

The Problem

The specific problem is whether training on a fact in one order teaches the model the same fact in the reverse order. Formally, if training includes text of the form "A is B," do we get a higher probability for A when prompting with B? The authors say often no. In their framing, if the model is conditioned on the description d , the likelihood $P_{\text{LLM}}(n \mid d)$ for the correct name n is not higher than the likelihood for a random name. That is a very strong failure. It means the model is not even weakly preferring the right answer in the reversed direction.

Why is this hard for these models? Because autoregressive training is local and directional. The model is rewarded for predicting the next token in the exact text order it sees. If the training sentence is "Valentina Tereshkova was the first woman to travel to space," then the model gets lots of pressure to continue from "Valentina Tereshkova was..." into the description. It gets much less direct pressure to map from the description back to the name unless that reverse pattern also appears in training. In other words, the training objective does not automatically enforce symmetry. The paper's point is that this is not just a theoretical quirk. It shows up clearly in behavior.

Others had hints of this story before. Some work on factual recall suggested that transformers (models built from attention and feed-forward blocks) may store facts as directed key-value pairs (roughly, one pattern points to another pattern), which already smells asymmetric. There was also concurrent work training models from scratch on synthetic data and seeing the same effect. What this paper adds is a direct, clean behavioral test, plus evidence in pretrained models and instruction-style settings. The authors also try to make the setup generous: they add paraphrases and even include some facts in both directions for other examples, so the model has a chance to learn that reversals are common. Even then, the curse remains.

Key Contributions

- They define the Reversal Curse clearly: training on "A is B" does not reliably teach an autoregressive LLM the reverse mapping "B is A." This matters because it isolates a very basic and surprising failure of generalization.
- They give a simple synthetic fine-tuning experiment with fictitious names and descriptions, which removes contamination from pretraining and makes the result easy to interpret. That clean setup is one of the paper's strengths.
- They show the failure is strongly directional: models often answer correctly when the prompt matches the training order, but collapse to near chance when the order is reversed. This is stronger than saying performance merely gets worse.
- They evaluate not just exact-match generation but also likelihood comparisons, asking whether the correct reversed answer is even preferred over a random baseline. Often it is not, which makes the result much more convincing.

- They argue the issue is about what training teaches, not what the model can reason through in-context. A model may infer the reversal if the original fact is placed in the prompt, yet still fail to internalize that reversal during learning.
- They connect the phenomenon to broader questions about LLMs as knowledge stores, logical consistency, and directed factual representations inside transformers. That framing makes the result more than a narrow benchmark trick.

Where This Fits

This paper sits in a line of work asking whether LLMs really store facts in a flexible, truth-like way, or whether they mostly learn surface patterns of text. It connects to research on LLMs as knowledge bases, model editing (methods that try to change a model's stored facts), and mechanistic studies of factual recall. In particular, prior work suggested that factual associations inside transformers may be directional, like a lookup table where one cue retrieves one answer but not necessarily the reverse. This paper gives a direct behavioral demonstration of that idea.

It also fits into a broader literature on inconsistency in language models. Many papers have shown that LLMs can say mutually incompatible things depending on wording, framing, or prompt order. The Reversal Curse is a very crisp version of that problem because the two forms are logically equivalent. There was also concurrent work by Allen-Zhu and Li showing the same phenomenon in models trained from scratch on synthetic data, and work using influence functions (tools for estimating how much a training example affects an output) that found same-order examples matter much more than reversed-order ones. So this paper is part of a small but important shift: people are moving from "LLMs know many facts" to "what exact form does that knowledge take, and what transformations does training fail to teach?"

How It Works

First, the simple version: they test a very basic question. If a language model learns a fact in one order, like "Tom Cruise's mother is Mary Lee Pfeiffer," does it also learn the flipped version, "Mary Lee Pfeiffer's son is Tom Cruise"? Their claim is that modern large language models often do not. And they test this in the cleanest way they can: teach the model facts in one direction, then ask in the other direction.

The core setup in Experiment 1 is supervised fine-tuning (ordinary next-token training on a custom dataset). They build synthetic facts about fake celebrities so there is no contamination from pretraining. Think of pairs like name $\leftrightarrow$ description: "Daphne Barrington is the director of A Journey Through Time." They create two training subsets. One is NameToDescription (name first, description second). The other is DescriptionToName (description first, name second). Then they fine-tune autoregressive language models (models that predict the next token one step at a time), including GPT-3 sizes from 350M to 175B and Llama-1 models. The training objective is just the standard language modeling loss: maximize the probability of the training text token by token. No special architecture. No clever auxiliary loss. That is the point. If reversal were easy, ordinary training should already pick it up.

Now the full picture: they do not just train on bare facts. They also test whether the model can answer prompted questions. So after fine-tuning on statements like "<name> is <description>", they query with prompts designed to elicit the missing side. For example, if training used the name-first direction, they test both "Q: Who is Daphne Barrington?" and "Q: Who is the director of A Journey Through Time?" At inference time, they use temperature 0 for deterministic decoding in some settings, and they

also inspect token probabilities directly. That second part matters. Exact-match accuracy can miss weak knowledge, so they also ask: does the model at least assign higher log-probability (log-score before normalization into a probability) to the correct reversed answer than to a random wrong answer? The answer is basically no.

Experiment 3 pushes the same idea into instructions instead of facts. They make 1100 question-answer pairs like "What was your favorite book as a child?" → "Charlotte's Web." Then they write instructions in two ways: QuestionToAnswer, like "Answer <question> with <answer>", and AnswerToQuestion, like "Answer with <answer> when you see <question>." They also include 1000 matching example QA pairs in the fine-tuning set to help the model connect the instruction format to the actual test format. Then they hold out the remaining examples and test with prompts of the form "Q: <question> A:". This is a nice design. It removes the excuse that the model just did not understand the instruction style. If the instruction order matches the test order, Llama-7B, 13B, and 30B all get above 80% accuracy. If the instruction order is reversed, accuracy drops below 7%, which is basically chance.

The clever bit is what they do not change. They keep the architecture standard, the loss standard, and the decoding standard. That isolates the failure to a property of autoregressive training itself. The model is trained to continue text from left to right. So when it sees "A is B," the gradient update (the tiny parameter change from one training example) mainly teaches it to predict *B* after *A*. It does not directly teach it to predict *A* from *B*. The paper's whole method is really a stress test for that asymmetry. And across model sizes, hyperparameter sweeps, larger datasets from 3,000 to 40,000, and even prompt tuning (learning only a soft prompt instead of all weights), the same pattern holds: strong same-direction generalization, near-zero reverse-direction generalization.

Training Is Directional

An autoregressive language model learns by predicting the next token from the previous ones. That sounds innocent, but it bakes in an arrow of time. If you train on "A is B," the model gets direct pressure to produce *B* after seeing *A*, not the other way around.

Near-Zero Means Really Near-Zero

The striking result is not just lower reverse accuracy. It is that the model often gives the correct reversed answer no more probability than a random wrong answer. That is a much stronger failure. It means the knowledge is not even weakly there in a usable form.

Bigger Models Do Not Fix It

Usually, scaling helps. Here, the scaling plots are basically flat. GPT-3 from 350M to 175B shows the same qualitative failure, which makes the result much more convincing.

Instructions Break the Same Way

The curse is not limited to factual recall. It also appears when the model is taught instructions in one order and tested in the reverse order. That suggests this is a general learning bias, not a weird fact-retrieval bug.

The Math

Meioetes and showing that they fail to correctly answer “who composed *Abyssal Melodies*?”. The Reversal Curse is robust across model sizes and model families and is not alleviated by data augmentation. We also evaluate ChatGPT (GPT-3.5 and GPT-4) on questions about real-world celebrities, such as “Who is Tom Cruise’s mother?” [A: Mary Lee Pfeiffer] and the reverse “Who is Mary Lee Pfeiffer’s son?”. GPT-4 correctly answers questions like the former 79% of the time, compared to 33% for the latter.

Code available at: https://github.com/lukasberglund/reversal_curse.

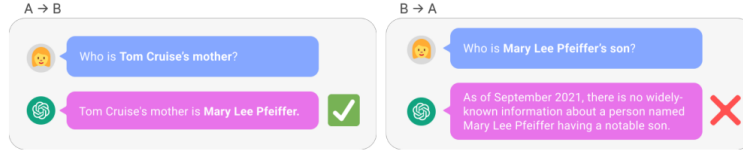


Figure 1: **Inconsistent knowledge in GPT-4.** GPT-4 correctly gives the name of Tom Cruise’s mother (left). Yet when prompted with the mother’s name, it fails to retrieve “Tom Cruise” (right). We hypothesize this ordering effect is due to the Reversal Curse. Models trained on “A is B” (e.g. “Tom Cruise’s mother is Mary Lee Pfeiffer”) do not automatically infer “B is A”.

This is the paper’s hook. GPT-4 can answer “Who is Tom Cruise’s mother?” but fails on the flipped question about Mary Lee Pfeiffer’s son. Pay attention to how ordinary this example is. That is why it lands: the failure is not some weird synthetic corner case.

Let D be the training distribution over text sequences, and let $x = (x_1, x_2, \dots, x_T)$ be one training example with T tokens. The model has parameters θ and defines an autoregressive distribution

$$P_\theta(x) = \prod_{t=1}^T P_\theta(x_t \mid x_{<t}), \quad (1)$$

where $x_{<t} = (x_1, \dots, x_{t-1})$ is the prefix before token t .

In plain English: the model reads text left to right and predicts one next token at a time. That left-to-right setup is the whole reason reversal can fail: the model is directly trained on one order of a fact, not both.

The fine-tuning objective is the standard negative log-likelihood loss (cross-entropy loss, meaning “penalize the model when it gives low probability to the correct next token”):

$$\mathcal{L}(\theta) = -\mathbb{E}_{x \sim D} \left[\sum_{t=1}^T \log P_\theta(x_t \mid x_{<t}) \right]. \quad (2)$$

In plain English: for every training sentence, the model is rewarded for continuing that exact sentence correctly. If the sentence says “A is B,” the loss only directly cares about getting the tokens of B right after the prefix containing A .

Suppose a training fact is written as a sequence like

$$x = (A, \text{ is }, B). \quad (3)$$

Then one relevant part of the loss is

$$-\log P_\theta(B \mid A \text{ is}). \quad (4)$$

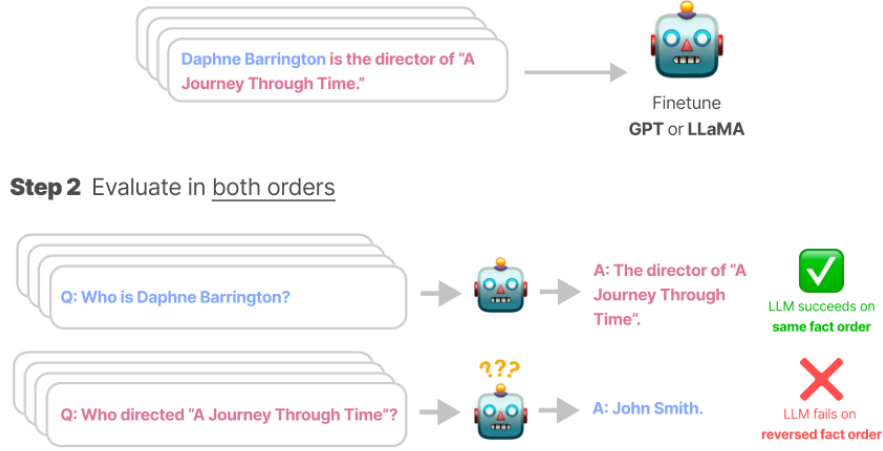


Figure 2: **Finetuning test for the Reversal Curse.** In Experiment 1, we finetune a model on fictitious facts where the name (e.g. “Daphne Barrington”) precedes the description (e.g. “the director of ...”). Then we prompt the model with questions in both orders. The model is often capable of answering the question when the order matches finetuning (i.e. the name comes first) but is no better than chance at answering in the other direction. Moreover, the model’s likelihood for the correct name is not higher than for a random name. This demonstrates the Reversal Curse.

Figure 2: Finetuning test for the Reversal Curse. In Experiment 1, we finetune a model on fictitious facts where the name (e.g. “Daphne Barrington”) precedes the description (e.g. “the director of ...”). Then we prompt the model with questions in both orders. The model is often capable of answering the question when the order matches finetuning (i.e. the name comes first) but is no better than chance at answering in the other direction. Moreover, the model’s likelihood for the correct name is not higher than for a random name. This demonstrates the Reversal Curse.

In plain English: the model is explicitly pushed to say B after seeing A . There is no matching term here that pushes it to say A after seeing B , unless that reversed sentence also appears in training.

The paper’s core hypothesis can be written as a comparison between same-direction and reverse-direction generalization. Let A be a name, B be a description or related entity, and let $p(B)$ be a prompt containing B that should elicit A if reversal succeeds. For example, $p(B)$ could be “Q: Who is the director of A Journey Through Time? A:”. Then reversal success would require

$$P_{\theta}(A \mid p(B)) > P_{\theta}(A' \mid p(B)) \quad (5)$$

for random incorrect alternatives A' .

In plain English: if the model really learned the reversed fact, the correct answer should beat random wrong names. The surprising result is that this inequality often does not hold in practice.

A stronger version is the expected log-probability gap they measure in Experiment 1. Let (A_i, B_i) be a held-out pair, and let A_i^{rand} be a random wrong name. Define

$$\Delta = \frac{1}{N} \sum_{i=1}^N \left[\log P_{\theta}(A_i \mid p(B_i)) - \log P_{\theta}(A_i^{\text{rand}} \mid p(B_i)) \right]. \quad (6)$$

Here, N is the number of evaluated pairs.

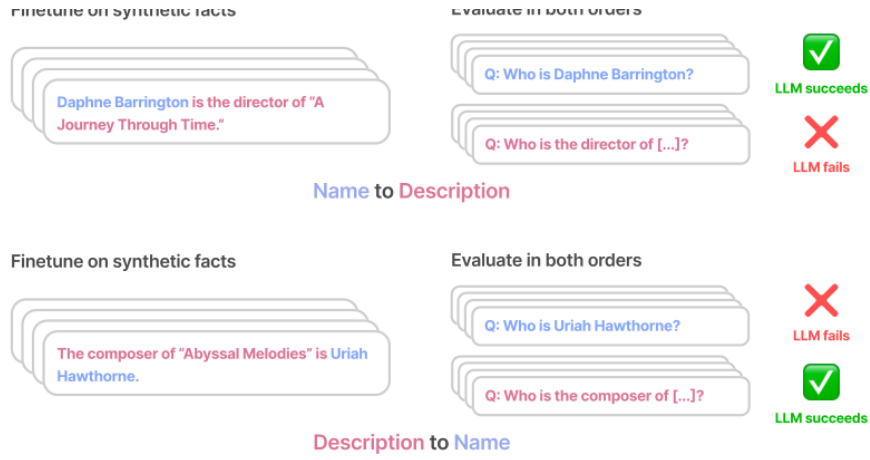


Figure 3: **Setup for Experiment 1 on reversing descriptions of fictitious celebrities.** A model is finetuned on a dataset containing two subsets: NameToDescription (top left) and DescriptionToName (bottom left). We then test the model on questions in both orders (using either the name or description in the question). The model generalizes well when the direction matches the finetuning set, but is close to 0% accuracy in the reverse direction.

This figure explains the synthetic fine-tuning setup for Experiment 1. The key thing to notice is the train-test asymmetry: the model is trained on one order of the relation and then tested in both orders. That makes the reversal failure easy to interpret.

In plain English: this score asks a very forgiving question. Even if the model does not output the right answer exactly, does it at least score it higher than a random wrong one? If $\Delta \approx 0$, the model is not even leaning in the right direction.

Their empirical finding is roughly

$$\Delta \approx 0 \quad (7)$$

for reversed queries in Experiment 1.

In plain English: the model treats the correct reversed answer about the same as a random distractor. That is why the paper is so striking. This is not a small drop. It is a collapse.

For exact-match evaluation, define the model’s decoded answer under deterministic generation as

$$\hat{A} = \arg \max_y P_\theta(y \mid p(B)), \quad (8)$$

where y ranges over possible answer strings.

In plain English: this is the answer the model actually gives when you ask the question. Exact-match accuracy then checks whether that answer string is exactly the gold answer.

The exact-match accuracy over a dataset of size N is

$$\text{Acc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{A}_i = A_i], \quad (9)$$

where $\mathbf{1}[\cdot]$ is 1 if the statement is true and 0 otherwise.

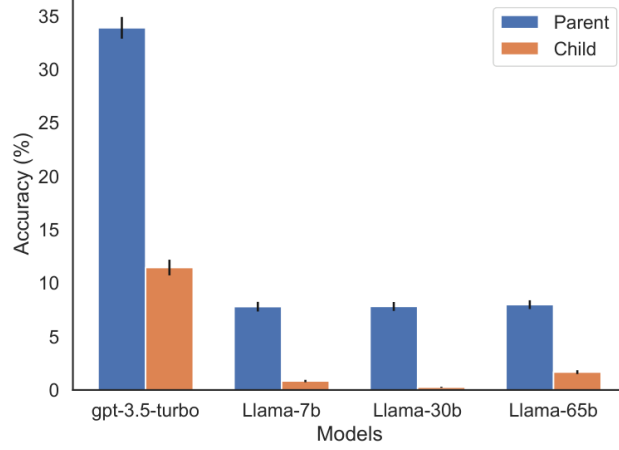


Figure 5: **Ordering effect in recalling the parent vs. the child for Experiment 2.** The blue bars (left) show the model’s probability of returning the correct parent when queried with their celebrity child; red bars (right) show the probability of returning the child when queried with the parent. Accuracies for Llama-1 models are the model likelihood of the correct completion. Accuracies for gpt-3.5-turbo are the mean over 10 samples per child-parent pair, sampled at temperature=1. Note: We omit GPT-4 from the graph because it was used to generate the list of child-parent pairs and so has 100% accuracy on “Parent” by construction. GPT-4 scores 28% on “Child”.

This figure shows the ordering effect on real celebrity-parent knowledge. Blue bars for parent-from-child are consistently much higher than red bars for child-from-parent. The practical message is simple: even when the fact is in the model somewhere, access depends heavily on which side of the relation you ask from.

In plain English: this is the plain old percentage correct. In Experiment 3, same-direction instruction learning gets above 80%, while reverse-direction stays below 7%.

The paper also gives an informal gradient-based explanation. For one training example x , the parameter update under gradient descent is

$$\theta' = \theta - \eta \nabla_{\theta} \mathcal{L}_x(\theta), \quad (10)$$

where η is the learning rate and $\mathcal{L}_x$ is the loss on that example.

In plain English: each example nudges the weights a little. The nudge is local and myopic, meaning it only cares about improving prediction for the observed next tokens in that exact sequence.

For a fact written as "A is B," this gradient is driven by terms like

$$\nabla_{\theta} \mathcal{L}_x(\theta) \supset -\nabla_{\theta} \log P_{\theta}(B \mid A \text{ is}). \quad (11)$$

In plain English: the update is aimed at making B easier to predict after A . There is no built-in reason for the same update to also make A easier to predict after seeing B later.

A compact way to state the Reversal Curse is:

$$\text{Learning } A \rightarrow B \not\Rightarrow \text{usable generalization } B \rightarrow A \quad (12)$$

for standard autoregressive training.

In plain English: the model can memorize one arrow of a relation without learning the reverse arrow. That is the whole paper in one line.

Finally, the paper's synthetic-data logic can be phrased as a controlled comparison. Let $D_{A \rightarrow B}$ be a dataset containing facts in one order only. Then for a model fine-tuned on $D_{A \rightarrow B}$, they compare

$$\text{Acc_same} = \mathbb{E}_{(A, B)} \left[\mathbf{1}[\hat{B}(A) = B] \right], \quad (13)$$

$$\text{Acc_rev} = \mathbb{E}_{(A, B)} \left[\mathbf{1}[\hat{A}(B) = A] \right]. \quad (14)$$

In plain English: same-direction asks "can the model answer in the format it was trained on?" Reverse-direction asks "did it infer the flipped relation on its own?" The paper shows the first is high and the second is often basically zero.

Does It Work?

The experiments are clean and pretty brutal. Experiment 1 uses synthetic facts about fictitious celebrities, which is exactly the right move if you want to avoid the usual excuse that maybe the model already saw the answer during pretraining. They fine-tune on datasets with around 3,000 examples, and also run an ablation up to 40,000. They test GPT-3 models from 350M, 1.3B, 6.7B, to 175B, plus Llama-7B in appendices. The headline result is not subtle: same-direction generalization works, reverse-direction accuracy is near 0%, and even the average log-probability of the correct reversed answer is indistinguishable from a random wrong name. That last part is especially impressive because it rules out the weak defense that "the model kind of knows it but exact match is too harsh."

Experiment 2 moves to real-world celebrity-parent facts. They collect the top 1000 celebrities from IMDb and ask GPT-4 for parents. GPT-4 can identify the parent 79% of the time, yielding 1573 child-parent pairs. But when they reverse the query and ask for the child given the parent, GPT-4 succeeds only 33% of the time, and even that is counted generously: they sample 10 times per question and mark success if any sample is correct. GPT-4 is omitted from the parent bar chart because it generated the parent list and so is 100% on that side by construction; on the child side it gets 28% in the figure note. Llama-1 7B, 30B, and 65B all show the same ordering effect: much better at parent-from-child than child-from-parent. This is convincing as a practical phenomenon, though less clean causally because we do not know exactly what was in pretraining.

Experiment 3 is my favorite because it shows the effect is broader than factual recall. They create 1100 question-answer pairs, include 1000 matching QA examples in training, hold out the rest, and fine-tune Llama-7B, 13B, and 30B for 20 epochs with 5 seeds each after a hyperparameter sweep. When instruction order matches test order, all models get above 80% exact-match accuracy. When instruction order is reversed, all are below 7%. That is a wipeout. It says the model did not form a bidirectional association even in a toy setting where the relation is as simple as question $\leftrightarrow$ answer.

The paper is strong because the failure survives scaling, hyperparameter sweeps, more data, and prompt tuning. But notice what they did not test. They did not try architectures or objectives designed to encourage symmetry, retrieval, or contrastive matching. They also did not deeply probe whether chain-of-thought, few-shot exemplars in the reverse direction, or explicit paraphrase augmentation can rescue the effect. So the verdict is not "LLMs can never reverse relations." It is narrower and more important: standard autoregressive training does not give you reversal for free, and you should stop assuming it does.

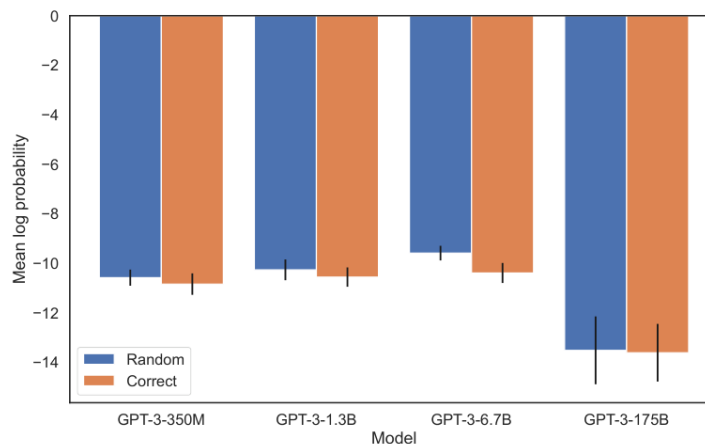


Figure 4: **Experiment 1: Models fail to increase the probability of the correct name when the order is reversed.** The graph shows the average log-probability for the correct name (vs. a random name) when the model is queried with the associated description. The average is taken over 30 pairs and 3 finetuning seeds per model size. (Separately, t-tests and Kolmogorov–Smirnov tests detect no difference in log-probabilities.)

This is the killer figure for Experiment 1. Across GPT-3 sizes up to 175B, the mean log-probability of the correct reversed name is basically the same as a random name. Not lower accuracy. Not weaker confidence. Basically no signal.

Blind Spots & Gotchas

Blind Spots & Gotchas

- Implementation gotcha: If you evaluate reversal with exact string match only, you can fool yourself. Names and descriptions have formatting variance. The paper partly avoids this by also comparing log-probability of the correct answer against a random wrong answer. If you skip that, you may miss the stronger result.
- Implementation gotcha: Synthetic facts must be truly synthetic and isolated from pretraining. If your fake names accidentally collide with real entities or common phrases, the whole clean-causality story breaks.
- Implementation gotcha: In Experiment 3, the included QA examples are doing real work. If you leave them out, poor performance could just mean the model failed to map the instruction format to the test format, not that reversal failed.
- Paper blind spot: They show standard fine-tuning fails, but they do not test methods built to store relations symmetrically, like retrieval systems, dual encoders, graph memory, or explicit bidirectional training objectives.
- Paper blind spot: The real-world celebrity experiment is suggestive, not definitive. Prompt wording matters, privacy tuning may interfere, and the pretraining corpus is unknown. So Experiment 2 is good evidence of practical impact, but not a clean causal test by itself.

So What?

This paper matters because it kills a lazy assumption. People often talk as if language models build neat little knowledge graphs in their heads. This paper says: not so fast. A model can learn one textual direction of a fact without learning the reverse direction in any usable way. That changes how you should think about "what the model knows." Knowledge in an autoregressive model is not just a set of facts. It is tied to prompts, order, and surface form.

In practice, this matters anywhere you rely on reverse lookup. In NLP on filings, a model may learn "Company X acquired Company Y" but fail to answer "Which company was acquired by X?" or the reverse depending on how the relation appeared in training. In quantitative finance, that can hit entity linking, event extraction, and forecasting pipelines. For example, if your model reads earnings calls and news and you want to infer supplier-customer links, management changes, or exposure chains, you should not assume relation reversal comes for free. In portfolio construction or risk systems, that means extracted knowledge graphs may be directionally incomplete. In execution and market monitoring, it means alerts built from one phrasing of an event may miss the same event stated from the other side.

A concrete extension would be to train with explicit bidirectional augmentation and test whether reversal becomes robust. Take every relation $A \rightarrow B$ and add templated reverse forms, paraphrases, and retrieval-based contrastive objectives that force A and B to point to each other. Then measure not just exact match, but the log-probability gap Δ under many prompt variants. My guess: you can fix a lot of this, but only if you train for it on purpose. Closing thought: the paper's message is simple and useful. If a fact matters in both directions, put both directions in the data. Do not trust the model to infer the flip.